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Clinical challenges in narcolepsy: From delayed diagnosis to daytime function impairments during treatment

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A R T I C L E I N F O

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A B S T R A C T

Objective: While a delayed diagnosis of narcolepsy is common, the reasons are not well understood. Additionally, only few systematic studies have focused on the daytime function of treated patients. In this cross-sectional online survey, we aimed to investigate the predictive factors for diagnostic delays and daytime dysfunction in Japanese patients with narcolepsy undergoing treatment.

Method: This survey included individuals aged 16 years or older with a confirmed diagnosis of narcolepsy. The participants completed a questionnaire related to disease diagnosis and current daytime function, including quality of life measured with Short Form-8 (SF-8) and work performance measured with the Work Productivity and Activity Impairment (WPAI) questionnaire.

Results: The study enrolled 248 of the 250 respondents. The mean time from symptom onset to definitive diagnosis was 8.7 years. Diagnostic delay was significantly associated with the year of symptom onset ($<$2005 vs. $\geq$2005), age at symptom onset ($<$16 vs. $\geq$16 years), narcolepsy type (without or with cataplexy), residence (rural vs. urban), and information sources about the disorder. The participants had a lower mental component summary score in SF-8 and higher WPAI absenteeism, presenteeism, and overall work impairment scores than the general Japanese population. Severe daytime sleepiness (Epworth Sleepiness Scale score $\geq$16) and younger age were associated with a poor mental component summary score and higher presenteeism and overall work impairment.

Conclusions: Improving sleepiness is crucial to optimize the patients' daytime functioning. A treatment system that addresses regional disparities in narcolepsy care is needed, along with increased public awareness, especially in educational settings.

1. Introduction

Narcolepsy is a sleep disorder characterized by frequent lapses during daytime and an increased propensity of rapid eye movement (REM) sleep. In addition to excessive daytime sleepiness (EDS), other cardinal symptoms of narcolepsy include cataplexy, sleep paralysis, and hypnagogic or hypnopompic hallucinations, as well as shallow and fragmented sleep [1–4]. These symptoms usually occur in adolescence or young adulthood, and a delay to definitive diagnosis is widely recognized as an important clinical issue. For example, previous studies conducted in Western countries and Japan reported a mean delay of 8–20 years [5–7]. Although several factors have been suggested to contribute to diagnostic delays, conclusive information based on comprehensive examinations is currently lacking.

Patients with narcolepsy often experience cognitive dysfunction, including attention deficits, memory impairment, and deficits in executive function, even with regular treatment [1]. These impairments may lead to reduced daytime function, which manifests as poor health-related quality of life (QOL) and work productivity, as shown previously [8–11]. However, no study has simultaneously assessed QOL and work productivity in the same patients with narcolepsy who receive regular treatment. Moreover, the factors associated with poor QOL or

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work productivity have not been comprehensively examined.

In light of these concerns, we conducted an online survey of patients with narcolepsy to investigate the following two clinical issues arising before a definitive diagnosis is issued and in the course of treatment: 1) level of diagnostic delay and associated factors, and 2) QOL and work productivity status and their associations with the symptoms that remain despite treatment.

2. Patients and methods

2.1. Patient selection and procedures

This cross-sectional online survey was conducted in accordance with the principles of the Declaration of Helsinki and the Clinical Trials Act of the Japanese Ministry of Health, Labour, and Welfare. The protocol was approved by the Ethics Review Committee of the non-profit organization "Japanese Association for the Promotion of State-of-the-Art in Medicine" (number, JPR40; approved on October 19, 2022). All participants provided informed consent before answering the questionnaire electronically.

Eligible patients were those who had already received a definitive diagnosis of narcolepsy and were receiving approved drugs in Japan, including modafinil, methylphenidate, and pemoline [1]. Patients were excluded from the study if they were younger than 16 years or had difficulty reading and writing in Japanese, which would hinder their ability to complete the questionnaire. The study population consisted of the following groups: 1) individuals who were members of the non-profit patient support group "Japan Narcolepsy Association; " 2) those who received prescribed drugs for narcolepsy from pharmacies belonging to the Nihon Chouzai Group; 3) those who were receiving medical care for the disorder from the investigators at specific medical institutions; and 4) those who received the prescribed drugs through the drug delivery service "TODOKUSURI," operated by the Japanese company Toppan Holdings Inc.

The patients who participated in the patient support group or used the drug delivery service were sent a survey invitation via postal mail or email, which included a Uniform Resource Locator (URL) to access the website. The patients who received the prescribed drugs from the pharmacy were invited to participate by the pharmacists. The survey invitation was also displayed at medical facilities in locations accessible to patients. Upon voluntary access to the survey website, the study participants provided informed consent electronically and responded to the questions in the questionnaire. All responses were based on the participants' self-reports.

2.2. Measures

The following information was collected from the participants as baseline characteristics: age, sex, place of residence by prefecture (i.e., administrative district in Japan), occupation (employer, employee, civil servant, self-employed, housewife or househusband, student, unemployed, or other), type of narcolepsy (presence or absence of cataplexy) [12], and previously suspected central nervous system disorders other than narcolepsy (insomnia, sleep apnea syndrome, circadian sleep–wake rhythm disorder, periodic limb movements during sleep, restless legs syndrome, insufficient sleep syndrome, developmental disorders such as attention-deficit hyperactivity disorder, depression/depressive state, epilepsy, orthostatic intolerance, anxiety disorder, adjustment disorder, or other). They also answered whether they had undergone polysomnography followed by a multiple sleep latency test (MSLT) for the diagnosis of narcolepsy.

The participants indicated the age when they first experienced EDS or cataplexy and when they received a definitive diagnosis of narcolepsy (both in years). This information allowed us to calculate the time interval from symptom onset to definitive diagnosis [7].

In addition, the participants were asked to report the number of institutions they had visited before receiving a definitive diagnosis. To explore the factors associated with the time from symptom onset to a definitive diagnosis, we inquired about the sources of information that the participants used to obtain knowledge about narcolepsy before visiting the medical institution where the diagnosis was confirmed. The sources were classified as follows based on previous studies [7,13,14]: television/radio, newspapers/magazines/books, general practitioners, the Internet/social networking services (SNS), or friends/acquaintances/family members. In addition, sex, physiological age at symptom onset, calendar year of symptom onset, type of narcolepsy (presence or absence of cataplexy), and place of residence (10 most densely populated prefectures vs. others) were also considered as candidate predictive factors for diagnostic delay in the statistical analysis. Age at symptom onset was classified into two categories ($<$16 vs. $\geq$16 years). As for the year of symptom onset, the median year reported by the participants (2005) was used as a cut-off point. The narcolepsy type was classified according to the International Classification of Sleep Disorders second edition (ICSD-2) [12] or first edition (ICSD-1) [15]. We did not use the terms "type 1" or "type 2," as defined in ICSD-3 [3], because we did not inquire whether the participants had received testing for orexin levels in the cerebrospinal fluid.

The participants were asked to individually report the frequency of the following symptoms at the time of the survey and before diagnosis: EDS, shallow and fragmented sleep, cataplexy, hypnagogic or hypnopompic hallucinations, and sleep paralysis. The frequency of each symptom was categorized as follows: at least once a day, at least once every 2 days, at least once every 3 days, at least once a week, at least twice a month, at least once a month, less than once a month, or never experiencing the symptom. The participants also rated the subjective inconvenience caused by each symptom before diagnosis and at the time of the survey. This level of inconvenience was self-rated on an 11-point Likert scale ranging from 0 to 10 [16].

QOL was measured using Short Form-8, and the physical component summary (PCS) and mental component summary (MCS) scores were calculated. Work productivity was measured using the Work Productivity and Activity Impairment (WPAI) questionnaire [17]. Based on the answers to the questionnaire, the following domain scores were calculated: absenteeism (percentage of work time missed due to health problems), presenteeism (percentage of impairment while working due to health problems), and overall work impairment (absenteeism plus presenteeism), with higher scores indicating lower productivity or more significant impairment. Furthermore, the participants self-rated their experiences using the Epworth Sleepiness Scale (ESS), a measure of subjective daytime sleepiness, and, depending on their scores, they were classified into two categories ($<$16 vs. $\geq$16) in the statistical analysis. Since a score of 16 or higher is associated with increased levels of daytime sleepiness, the threshold was set at an ESS score of 16 [18].

2.3. Statistical analyses

Baseline characteristics were descriptively summarized as means with standard deviation (SD) for continuous variables and frequencies with percentages for categorical variables. The frequency of EDS was classified as at least once every 2 days or less, whereas the frequency of cataplectic episodes was classified as at least once a week or less, based on the Swiss Narcolepsy Scale [19]. The frequencies of other symptoms were also classified as at least once a week or less.

T-tests were used to compare the mean PCS and MCS scores between each pair among the following groups: participants with cataplexy, those without cataplexy, and the general Japanese population. The mean domain scores of WPAI were also compared in the same manner using the values from the general Japanese population reported in a previous study [20]. Post hoc analyses were performed to compare the scores three times, with p-values adjusted using the Bonferroni method. To examine the impact of demographic factors on the results, the same comparisons were conducted using analysis of variance (ANOVA) adjusted for sex and age. The Pearson's product-moment correlation coefficient was also calculated in the post-hoc analysis between each PCS and MCS score and each domain score in WPAI.

Differences in the mean levels of disease-related inconveniences were compared using t-tests. Comparisons were also performed using ANOVA, considering sex and age. Additionally, Spearman's rank correlation coefficient was employed to assess the correlation between the levels of inconvenience before diagnosis and at the time of the survey. In these analyses, the Bonferroni method was used for multiplicity adjustment.

Furthermore, Poisson regression analysis was performed to explore the factors associated with the time from symptom onset to definitive diagnosis. The explanatory variables included in the model (i.e., potential predictive factors) have already been described above. Multivariable linear regression analysis was also performed to explore the factors associated with QOL or WPAI (scores for absenteeism, presenteeism, and overall work impairment). The following explanatory variables were included in the model: age at the time of the survey, sex, employment status (employed full-time vs. employed part-time or unemployed), type of narcolepsy, time from symptom onset to definitive diagnosis, and ESS score (<16 vs. ≥ 16). Age at the time of the survey and time from symptom onset to definitive diagnosis were treated as continuous variables.

The data were analyzed using the statistical analysis software R version 4.3.0 (R Foundation for Statistical Computing, Vienna, Austria).

3. Results

3.1. Disposition of participants

Between November 11, 2022 and June 30, 2023, 250 participants met the eligibility criteria. Of these, 248 were included in the study after excluding two participants due to uncertainty regarding the presence or absence of cataplexy as well as for receiving a diagnosis after the introduction of ICSD-2 [12] in Japan in 2010. Since then, MSLT has become an essential tool for diagnosing narcolepsy without cataplexy. The participants completed the questionnaire within the appropriate time frame—mean time ±2 SD, which is generally applied to web surveys in Japan to exclude unfaithful answers completed in too short or too long a time period; abnormal answers such as repeating an answer to different questions or checking all potential answers to the same question were not observed.

3.2. Baseline characteristics

Table 1 presents the baseline characteristics of the participants. More than half of the participants (n = 138, 55.6 %) were women. The mean (SD) age at the time of the survey was 35.6 (13.4) years, and the age of diagnosis was 24.8 (10.3) years. Over half of the participants (n = 143, 57.7 %) experienced cataplexy. The remaining 105 (42.3 %) had no such experience (n = 100) or did not know whether they had cataplexy (n = 5). All participants had received pharmaceutical therapy; of these, 199 (80.2 %) received modafinil. Previously suspected central nervous system disorders other than narcolepsy in the study participants are listed in Supplementary Table 1.

Supplementary Table 2 presents the frequency of typical narcolepsy-related symptoms in the cohort. Before diagnosis, 242 participants (97.6 %) experienced EDS at least once every 2 days, and more than 40 % of the participants reported the following symptoms at least once a week: shallow and fragmented sleep (62.9 %), cataplexy (45.2 %), hypnagogic or hypnopompic hallucinations (57.3 %), or sleep paralysis (41.1 %). At the time of the survey, 166 participants (66.9 %) had EDS at least once every 2 days. The percentage of participants who experienced cataplexy, hypnagogic or hypnopompic hallucinations, and sleep paralysis at least once a week was low, at approximately 30 % or less. In contrast, more than half of the participants (55.6 %) reported shallow and fragmented sleep at least once a week. The mean (SD) ESS score at the time of the survey was 16.9 (4.9), with 87.1 % and 66.9 % of the participants having ESS scores of at least 11 and 16, respectively, despite receiving treatment for the disorder. Only 12.9 % had a score within the normal range (≤ 10).

Table 1
Descriptive variables of the study participants.

	Overall	with CA	without CA
	n = 248	n = 143	n = 105
Sex, n (%)			
Male	110 (44.4)	65 (45.5)	45 (42.9)
Female	138 (55.6)	78 (54.5)	60 (57.1)
Age (years)			
Mean (SD)	35.6 ± 13.4	37.8 ± 14.3	32.8 ± 11.8
Age at onset t of narcoleptic symptom (years)			
Mean (SD)	16.1 ± 8.5	16.4 ± 9.8	15.6 ± 6.3
Age of receiving definite diagnosis (years)			
Mean (SD)	24.8 ± 10.3	25.1 ± 11.3	24.4 ± 8.9
Type of Narcolepsy, n (%)			
NA with CA	143 (57.7)	143 (57.7)	–
NA without CA	100 (40.3)	–	100 (40.3)
Unknown[a]	5 (2.0)	–	5 (2.0)
Type of Medications, n (%)			
Modafinil	199 (80.2)	116 (81.1)	83 (79.0)
Methylphenidate	97 (39.1)	63 (44.1)	34 (32.4)
Pemoline	28 (11.3)	15 (10.5)	13 (12.4)
Antidepressants	84 (33.9)	73 (51.0)	11 (10.5)
Other psychotropic medications	7 (2.8)	3 (2.1)	4 (3.8)
Sleep medications	40 (16.1)	27 (18.9)	13 (12.4)
Other	25 (10.1)	17 (11.9)	8 (7.6)
No prescription drugs	0 (0)	0 (0)	0 (0)
Occupation, n (%)			
Employed full time	187 (75.4)	110 (76.9)	77 (73.3)
Employed part time or unemployed	61 (24.6)	33 (23.1)	28 (26.7)
Prefecture of Residence, n (%)[b]			
Prefectures with Population Density in the Top 10 or Higher	185 (74.6)	103 (72.0)	82 (78.1)
Prefectures with Population Density in the Top 11 or Lower	63 (25.4)	40 (28.0)	23 (21.9)

Abbreviations: NA, narcolepsy; CA, cataplexy; SD, standard deviation.
[a] There were five cases where the presence or absence of CA was unknown, but these were included in the without CA group for analysis.
[b] The top 10 prefectures by population density are Saitama, Chiba, Tokyo, Kanagawa, Aichi, Kyoto, Osaka, Hyogo, Fukuoka, and Okinawa.

3.3. Time from symptom onset to definitive diagnosis

The mean (SD) time from symptom onset to definitive diagnosis was 8.7 (8.2) years. The mean (SD) number of medical institutions visited before a definitive diagnosis was 1.5 (1.3), which did not include the institution where narcolepsy was finally diagnosed. Regarding the information sources, 64.9 % of the participants used the Internet/SNS to obtain knowledge about narcolepsy before visiting a medical institution where narcolepsy was diagnosed, whereas 13.3 % used newspapers/magazines/books, and 8.9 % used television/radio. The remaining 26.2 % and 18.5 % obtained information from friends/acquaintances/family members and from the attending general practitioners, respectively.

Table 2 presents the results of the Poisson regression analysis. The year of symptom onset ($\geq$2005) was negatively associated with the time to receive a definitive diagnosis, with a risk ratio of 0.360 (95 % confidence interval [CI], 0.325 to 0.398; $p < 0.001$). The following factors

Table 2
Multivariate analyses of factors associated with time to the definitive diagnosis.

Variables	Risk ratio	95% CI lower	95% CI upper	p value
Female (Ref. Male)	1.072	0.983	1.169	0.117
Narcolepsy with Cataplexy (Ref. without CA or unknown)	0.900	0.825	0.983	0.019
Age at onset of subjective symptom (Ref. ≥ 16 years old)	0.832	0.754	0.918	0.000
Year of onset (Ref. ≥ 2005)	0.360	0.325	0.398	0.000
Place of Residence (Ref. Prefectures outside the 10 most densely populated)	0.834	0.758	0.917	0.000
Sources of information used to learn about narcolepsy before diagnosis				
TV or Radio (Ref. not used)	1.263	1.107	1.442	0.001
Newspaper, magazine or book (Ref. not used)	1.198	1.072	1.339	0.001
Primary Care Physician (Ref. not used)	0.945	0.834	1.071	0.377
Internet or SNS (Ref. not used)	1.208	1.097	1.330	0.000
Friend, acquaintance or family member (Ref. not used)	0.865	0.777	0.962	0.008

Abbreviations: SNS, social networking services.

were also negatively associated with time to diagnosis: narcolepsy with cataplexy ($p = 0.019$), age at symptom onset (≥16 years), and place of residence (10 most densely populated prefectures) ($p < 0.001$ for both). In addition, receiving information from friends/acquaintances/family members was negatively associated with time to diagnosis ($p = 0.008$). In contrast, sourcing information from television/radio, newspapers/magazines/books, and the Internet/SNS was positively associated ($p = 0.001$, $p = 0.001$, and $p < 0.001$, respectively).

3.4. Level of subjective inconvenience due to narcoleptic symptoms

The level of subjective inconvenience caused by EDS was significantly greater than that caused by any other symptom, both before diagnosis and at the time of the survey ($p < 0.001$ for each comparison) (Fig. 1). Furthermore, the level of subjective inconvenience due to narcoleptic symptoms was assessed separately for individuals with cataplexy and those without cataplexy. Consistent with the overall analysis results, the subjective inconvenience caused by EDS was significantly pronounced in both groups compared to that due to other symptoms, both prior to diagnosis and at the time of the survey ($p < 0.001$ for each comparison) (Supplementary Fig. 1). The participants reported being less troubled by symptoms after diagnosis. The level of inconvenience caused by narcoleptic symptoms before diagnosis significantly correlated with that at the time of the survey ($p < 0.01$ for each symptom).

3.5. QOL and work productivity

The mean MCS scores of participants with and without cataplexy were significantly lower than that in the general Japanese population (43.6 vs. 49.3, adjusted $p < 0.001$; 43.4 vs. 49.3, adjusted $p < 0.001$, respectively). In contrast, their mean PCS scores did not differ from that of the general population (48.2 vs. 48.9, adjusted $p = 0.724$; 49.8 vs. 48.9, adjusted $p = 0.702$, respectively) (Fig. 2). The mean domain scores in WPAI were significantly higher in the two participant subgroups than in the general Japanese population (absenteeism: 7.8 vs. 2.3, adjusted $p = 0.018$, and 6.1 vs. 2.3, adjusted $p = 0.091$, respectively; presenteeism: 49.5 vs. 17.7, adjusted $p < 0.001$, and 41.1 vs. 17.7, adjusted $p < 0.001$, respectively; and overall work impairment: 51.7 vs. 19.0, adjusted $p < 0.001$, and 42.9 vs. 19.0, adjusted $p < 0.001$, respectively) (Fig. 3). Additionally, in patients with narcolepsy, both the PCS and MCS scores correlated significantly with all WPAI domain scores (Supplementary Table 3).

Table 3 presents the results of the multivariable regression analysis. This analysis did not target PCS because the mean scores did not differ between each participant subgroup and the general Japanese population. The MCS score negatively correlated with the ESS score (≥16) (regression coefficient, −2.441 [95 % CI −4.737 to −0.144], $p = 0.038$), and positively correlated with age at the survey (regression coefficient, 0.109 [95 % CI, 0.019 to 0.198], $p = 0.018$). As for the WPAI domain scores, higher presenteeism and overall work impairment were

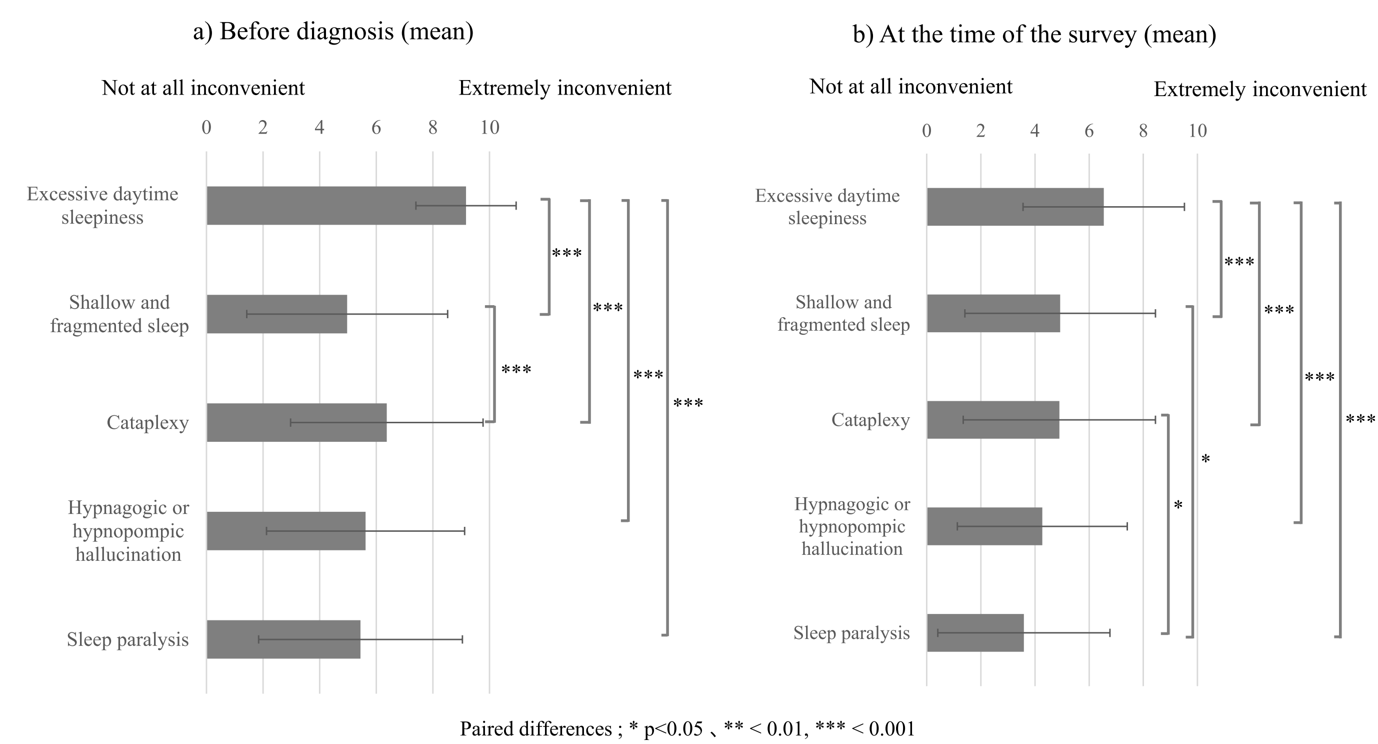

Fig. 1. Level of inconvenience caused by each symptom of narcolepsy a) before diagnosis and b) at the time of the survey
The mean values were compared using the *t*-test.

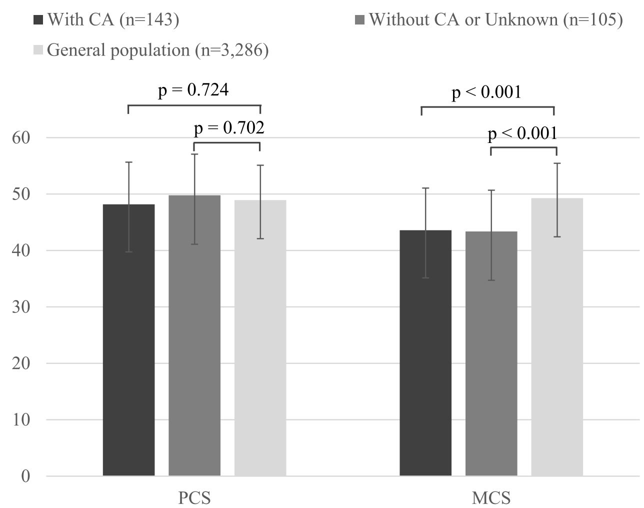

Fig. 2. Comparisons of the physical component summary and mental component summary scores of Short Form-8 between each pair of participants with cataplexy (n = 143), participants without cataplexy or those who did not know the type of narcolepsy (n = 105), and the general Japanese population (n = 3286)
The data are expressed as the mean ± standard deviation. Mean values were compared using the *t*-test. The P-values were adjusted using the Bonferroni method. SF-8, Short Form-8; PCS, physical component summary; MCS, mental component summary; CA, cataplexy.

significantly associated with a higher ESS score (≥16) and younger age.

4. Discussion

This is the first study to comprehensively examine the factors that contribute to a delayed definitive diagnosis of narcolepsy and the level of daytime function when receiving treatment, as well as aggravating factors, in the same patients. We observed a significant delay in diagnosing narcolepsy, with a mean time from symptom onset to definitive diagnosis of nearly 9 years, consistent with that in previous reports [5–7]. Additionally, patients with idiopathic hypersomnia typically face an average diagnostic delay of 10 years [21]. Given these observations, a tendency for delayed diagnosis across central hypersomnia as a whole becomes evident.

Several factors were associated with the diagnostic delay, including year of symptom onset (<2005 vs. ≥2005), as reported previously [6,7]. This finding may be related to the insufficient dissemination of the concept of narcolepsy and lack of medical facilities that could properly diagnose and treat this disorder in the past. Age at symptom onset (<16 vs. ≥16 years) was also associated with diagnostic delays, suggesting that younger age at symptom onset results in a longer time to diagnosis. This result is in agreement with that of a previous European study in patients with narcolepsy type 1, which found that a younger age at symptom onset was associated with diagnostic delays [5]. The school-age years are crucial for psychosocial development, making it important to diagnose and treat narcolepsy as early as possible. To support this notion, schools—especially junior high schools—should implement activities that raise awareness about the symptoms of the disorder. Notably, even when multivariate analysis was conducted using age at symptom onset and year of symptom onset as continuous variables, similar results were observed, indicating that younger age at onset and older generational cohorts are associated with a longer time to diagnosis. The changes in sleep patterns among adolescents, as well as stress and fatigue unique to this developmental stage, may also be associated with daytime sleepiness [22,23]. To prevent the inappropriate use of psychostimulant medications stemming from misdiagnosis, it is essential to raise awareness in educational settings, together with narcolepsy and the general causes of excessive sleepiness. Furthermore, physicians should thoroughly check for the presence of issues in their patients' lifestyles that may contribute to daytime sleepiness before making diagnoses.

On the other hand, experiences of cataplexy were found to be negatively associated with diagnostic delay. This suggests that narcolepsy without cataplexy—the most characteristic symptom of the disorder—may be easily overlooked. Therefore, it is vital to raise awareness about narcolepsy without cataplexy, which is now referred to as "narcolepsy type 2," both among the general population and in clinical practice. Notably, EDS occurrence may depend on various sleep hygiene issues such as sleep deprivation and circadian rhythm dysregulation, and the MSLT may yield false-positive results for narcolepsy type 2. These observations indicate that particular caution is required when diagnosing a type 2 disorder [24,25].

Of note, residence in the 10 most densely populated prefectures was negatively associated with diagnostic delay. In Japan, physicians and institutions certified for sleep disorders are concentrated in large cities

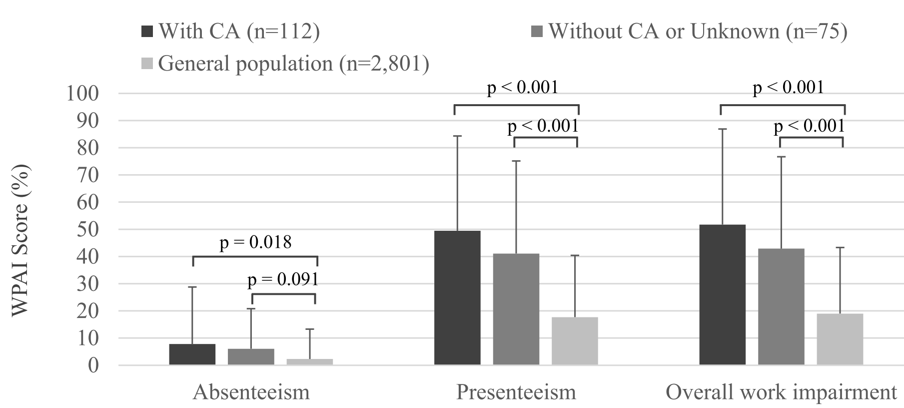

Fig. 3. Comparison of Work Productivity and Activity Impairment domain scores between each pair of participants with narcolepsy (n = 112), participants without cataplexy or those who did not know the type of narcolepsy (n = 75), and the general Japanese population (n = 2801)
The data are expressed as the mean ± standard deviation. Mean values were compared using the *t*-test. The P-values were adjusted using the Bonferroni method. WPAI, Work Productivity and Activity Impairment; Absenteeism, percentage of work time missed due to health problems; Presenteeism, percentage of impairment while working due to health problems; Overall work impairment, absenteeism plus presenteeism; CA, cataplexy.

Table 3
Multivariate linear regression analysis of factors based on the MCS and WPAI scores.

Variables[a]	MCS (SF-8) Partial regression coefficients [95% CI]	p-value	Absenteeism Partial regression coefficients [95% CI]	p-value	Presenteeism Partial regression coefficients [95% CI]	p-value	Overall work impairment Partial regression coefficients [95% CI]	p-value
Narcolepsy with Cataplexy (Ref. without CA or unknown)	0.101 [-2.105, 2.306]	0.929	1.468 [-4.308, 7.244]	0.619	6.615 [-3.627, 16.858]	0.207	6.952 [-3.365, 17.269]	0.188
ESS ≥16 (Ref. ESS<16)	−2.441 [-4.737, −0.144]	0.038	2.873 [-3.319, 9.066]	0.364	16.069 [5.090, 27.049]	0.005	16.109 [5.049, 27.168]	0.005
Time to diagnosis (years)	−0.015 [-0.157, 0.126]	0.831	0.198 [-0.168, 0.565]	0.290	−0.168 [-0.817, 0.482]	0.614	−0.016 [-0.670, 0.639]	0.963
Age at the survey	0.109 [0.019, 0.198]	0.018	−0.096 [-0.350, 0.157]	0.458	−0.564 [-1.013, −0.114]	0.015	−0.536 [-0.989, −0.083]	0.022
Female (Ref. Male)	−1.384 [-3.519, 0.750]	0.205	1.698 [-3.822, 7.218]	0.547	−0.151 [-9.939, 9.637]	0.976	−0.118 [-9.978, 9.741]	0.981
Employed full time (Ref. Employed part time or Unemployed)	−0.391 [-2.873, 2.090]	0.757	−2.080 [-11.307, 7.147]	0.659	6.385 [-9.976, 22.746]	0.445	4.054 [-12.426, 20.535]	0.630

Abbreviation: ESS, Epworth Sleepiness Scale.
[a] Time to diagnosis and age were treated as continuous variables, while all other variables were treated as categorical variables.

[26]. If this trend is associated with diagnostic delays, a system allowing collaboration between rural general practitioners and urban sleep specialists should be established. As for the information sources, the use of television/radio, newspapers/magazines/books, and the Internet/SNS was positively associated with a delay in receiving a diagnosis. Although the reason is unclear, this result is in line with a previous review finding, which showed that the use of the Internet and SNS did not seem to be effective in motivating patients to seek medical attention [27]. In contrast, information obtained from friends, acquaintances, and family members was linked to a shorter diagnostic delay. This suggests that obtaining insights from individuals who deeply understand a patient's condition on a daily basis can help the patient obtain a definitive diagnosis more quickly. If so, efforts to increase knowledge related to narcolepsy among the general population are necessary to promote early diagnosis. The mass media and the Internet should disseminate necessary and accurate disease-related information to achieve this goal.

The mean ESS score at the time of the survey was 16.9, and only 12.9 % of the participants had an ESS score within the normal range (≤10). The results indicate that most participants continued to experience EDS despite receiving treatment with psychostimulants, primarily modafinil, suggesting that administration of modafinil as a primary treatment may not sufficiently reduce daytime sleepiness in many patients. A similar trend was observed in the clinical trials of modafinil for narcolepsy [28,29]. Furthermore, the level of inconvenience caused by EDS was significantly greater than that caused by any other narcoleptic symptom. It is essential to investigate this issue with newer drugs for narcolepsy treatment such as pitolisant, sodium oxybate, and solriamfetol [30], none of which is currently approved in Japan.

At the time of the survey, the mean MCS score of participants with or without cataplexy was significantly lower than that of the general Japanese population, and a lower MCS score was significantly associated with a higher ESS score, consistent with our previous study results [9,10]. Additionally, all WPAI domain scores were significantly higher than those of the general Japanese population, and an ESS score of 16 or more was significantly associated with higher presenteeism and overall work impairment, in agreement with the results by Bassi et al. [11]. However, different from their study, we included patients with narcolepsy without cataplexy, who were assumed to have type 2 narcolepsy. Consequently, although the subtype of narcolepsy was not relevant, we found that higher levels of EDS, as indicated by the ESS score, were associated with a lower MCS score as well as higher presenteeism and overall work impairment among the participants.

Interestingly, considering that trends in the changes in measures of daytime function demonstrated significant correlations, incorporating scheduled naps [31], adjusting medication, and improving sleep habits should be considered in cases of EDS despite treatment. Additionally, younger age was significantly associated with higher presenteeism and overall work impairment. However, given the generally increased presenteeism in younger individuals [32,33], age-specific work impairment may not be evident in patients with narcolepsy.

Several limitations should be mentioned. First, some participants might have had hypersomnia disorders other than narcolepsy because the diagnosis relied on self-reported information. Furthermore, all patients diagnosed after 2010, when MSLT became mandatory for diagnosing narcolepsy, underwent MSLT if they did not exhibit cataplexy. In contrast, the diagnosis of many patients before 2010 (n = 96) was primarily based on the ICSD-1 criteria. Additionally, some of these patients did not undergo MSLT, specifically 31 who had cataplexy and 10 who either did not have cataplexy or had or had an unknown status. Consequently, the diagnoses for some participants identified before 2010 may not be accurate. In five patients, the presence or absence of cataplexy was unknown, but these individuals were diagnosed using MSLT, and their data were included in the group without cataplexy for analysis. Furthermore, our participants included individuals who joined a patient support group, those who were visiting medical institutions, and those who received prescribed drugs at the pharmacy or via a drug delivery service. Consequently, patient characteristics may vary among these groups. Moreover, the participants in this study could use the Internet without difficulty. Therefore, our results should be interpreted cautiously because of methodological issues in web surveys, such as under-coverage and self-selection [34]. Given these limitations, we should carefully consider the extent to which our results can be applied to patients with narcolepsy in general. Second, some participants may have not accurately recalled their past conditions such as age at symptom onset or daily life situations. It is important to note that the average age of the participants in this study was 35.6 ± 13.4 years, while the average age at onset was 16.1 ± 8.5 years. This means that the information regarding the onset was recalled from approximately 20 years ago. Therefore, the potential impact of significant recall bias should be taken into consideration. Third, we did not investigate the influence of sleep duration, concomitant diseases, treatment adherence, or adverse reactions to the prescribed drugs, including psychotropic agents. Finally, although psychosocial factors and levels of depression and anxiety have been reported to affect QOL and the work productivity of patients with narcolepsy [9,11], we did not investigate these factors.

5. Conclusions

In this cross-sectional online survey of patients with narcolepsy, we found a substantial diagnostic delay that was influenced by various factors. To expedite this process, general practitioners in rural areas must facilitate referrals to sleep specialists. Additionally, improving the quality of information about narcolepsy provided by the mass media can better support patients and their families in seeking medical help. It is also vital to ensure that accurate information about narcolepsy is widely available in society. Furthermore, patients reported a considerable decline in both QOL and work performance, even while receiving treatment. Notably, EDS was associated with a poor mental health QOL score as well as higher presenteeism and overall work impairment. Identifying this issue early and providing comprehensive care, including enhancement of the work environment, should become mandatory. Future prospective studies should examine how early diagnosis and comprehensive care can improve the prognosis for patients with narcolepsy.

CRediT authorship contribution statement

Yuichi Inoue: Writing – review & editing, Validation, Supervision, Resources, Methodology, Investigation, Conceptualization. **Takuya Kumagai:** Writing – review & editing, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization. **Ayako Shoji:** Writing – original draft, Visualization, Validation, Software, Project administration, Investigation, Formal analysis, Data curation. **Kinya Kokubo:** Writing – review & editing, Methodology. **Makoto Honda:** Writing – review & editing, Validation, Supervision, Resources, Methodology, Investigation, Conceptualization.

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Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Takuya Kumagai is employee of Aculys Pharma, Inc. Healthcare Consulting, Inc. has received research funding from Aculys Pharma, Inc. Kinya Kokubo is the representative director of Healthcare Consulting, Inc. Ayako Shoji is employee of Healthcare Consulting, Inc. Yuichi Inoue and Makoto Honda has received advisory fees from Aculys Pharma, Inc. for this research. Beyond these declared interests, there are no conflicts of interest to disclose regarding this study.

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Glossary

EDS, excessive daytime sleepiness; ESS, Epworth Sleepiness Scale; ICSD, International Classification of Sleep Disorders; MCS, mental component summary; PCS, physical component summary; QOL, quality of life; WPAI, Work Productivity and Activity Impairment.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.sleep.2025.106496.

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